COLLEGE BUSINESS EDUCATION

DAR ES SALAAM

GROUP ASSIGNMENT

COURSE: BACHELOR DEGREE IN INFORMATION TECHINOLOGY.

MODULE: DATA STRUCTURES AND ALGORITHMS

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FACILITATOR: AHMED KIJAZI

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GROUP NUMBER THREE.

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Question: With examples, explain different types of tree data structures

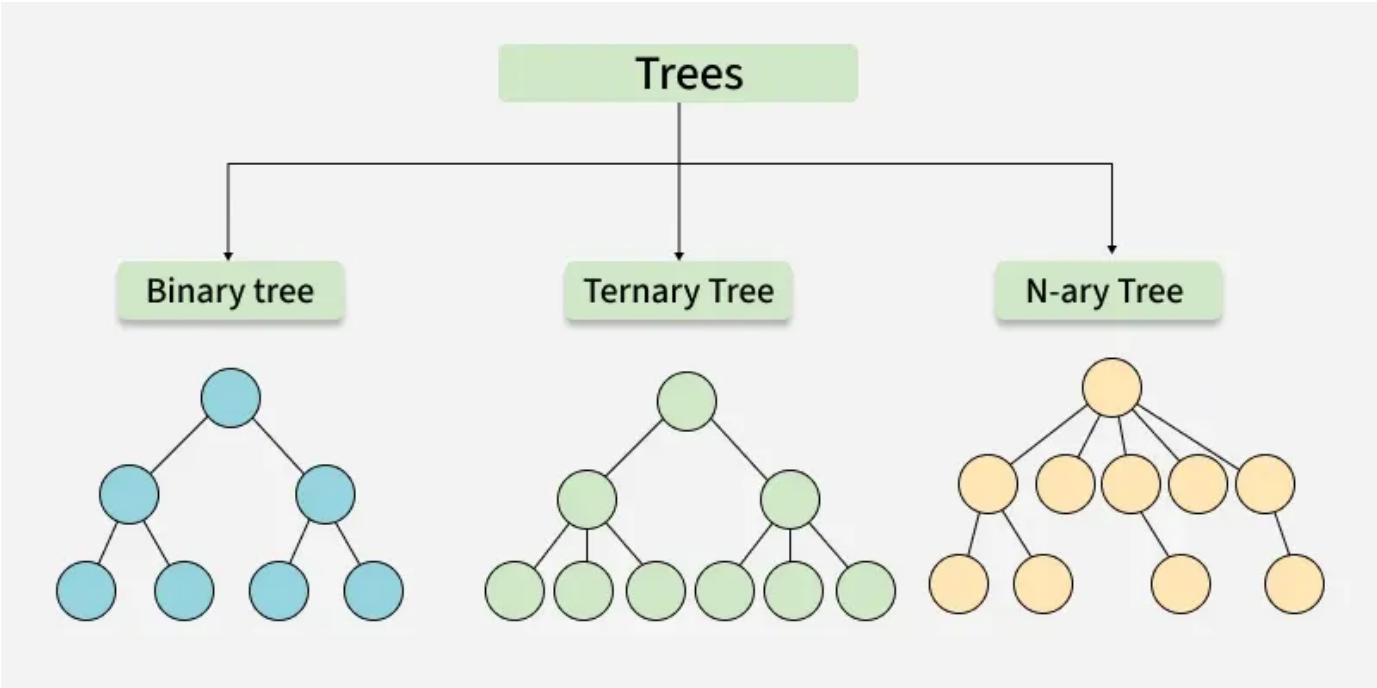
Answers

A Tree is a hierarchical data structure that consists of nodes connected by edges. It is used to represent relationships between elements, where each node holds data and is connected to other nodes with edges.

There are three Types of Tree data Structures

- Binary Tree
- Ternary Tree
- N-ary Tree

Consider a diagram below



i. Binary Tree

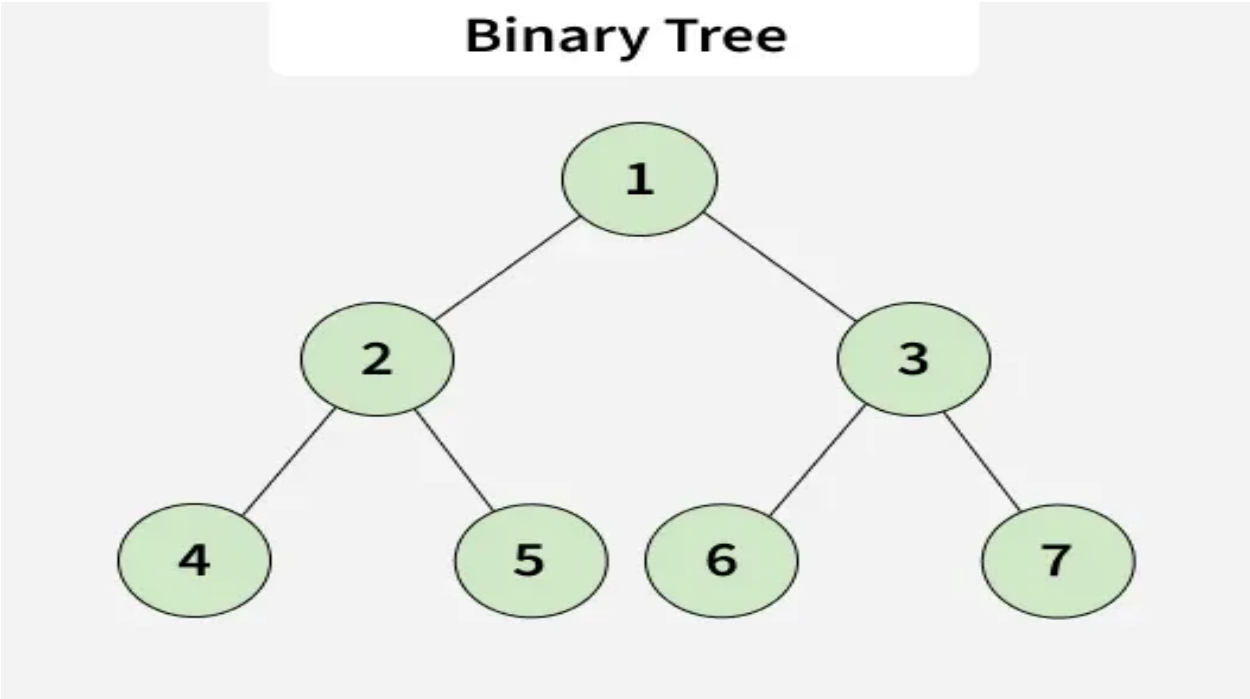
A binary tree is a tree data structure where each node has at most two children. These two children are usually referred to as the left child and right child.

Examples of Binary tree

- **Full Binary Tree** – every node has either 0 or 2 children.
- **Complete Binary Tree** – all levels are fully filled except possibly the last, which is filled from left to right.
- **Balanced Binary Tree** – height difference between left and right subtrees of every node is minimal

In Binary tree each node can have a maximum of two children.

Consider the tree below



ii. Ternary Tree

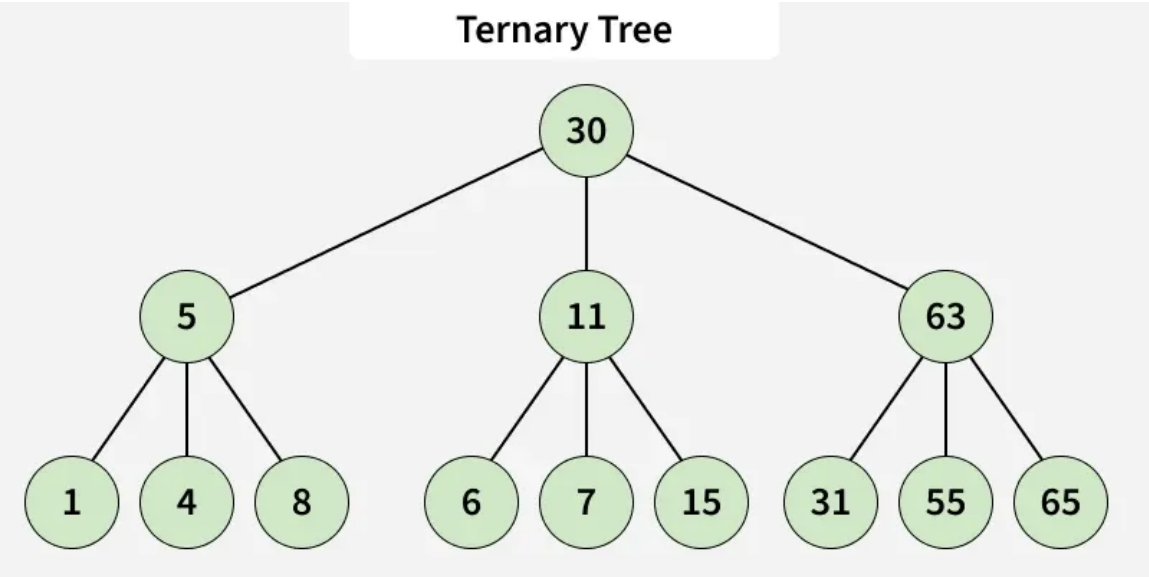
A Ternary Tree is a tree data structure in which each node has at most three children. These two children are usually referred to as the left child, mid child and right child.

Examples of Ternary Tree:

- **Ternary Search Tree**: A ternary search tree is a special trie data structure where the child nodes of a standard trie are ordered as a binary search tree.

- **Ternary Heap:** A type of heap where each node can have up to three children, though less common than binary heaps

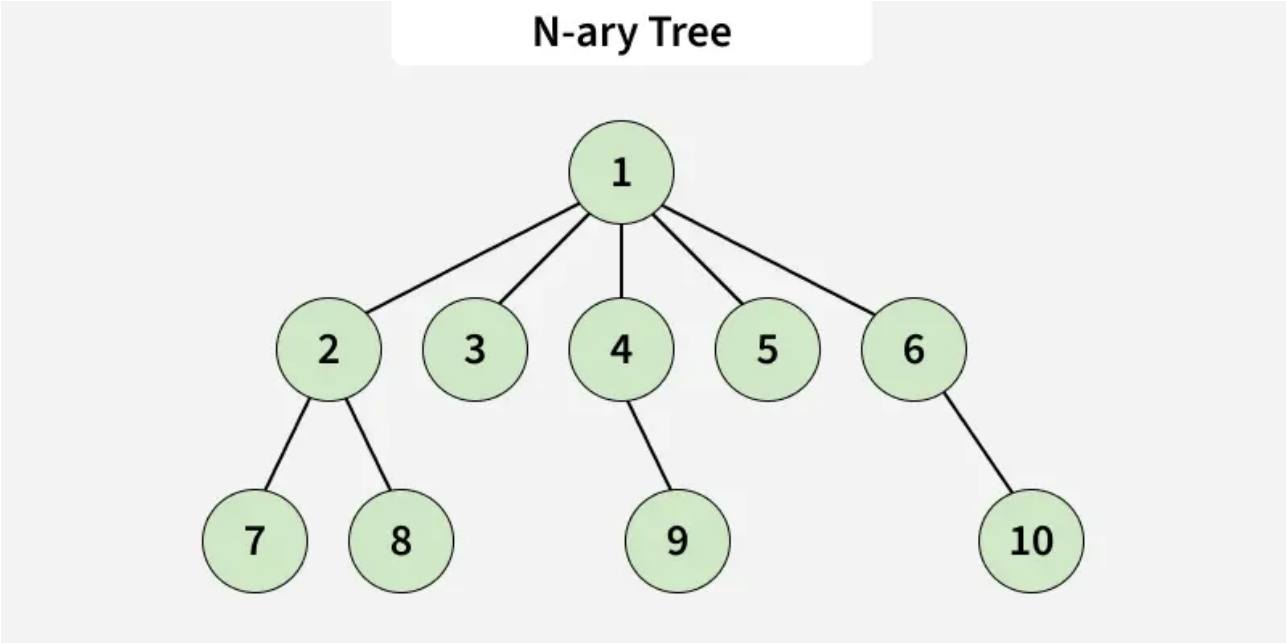
Consider the tree below.



iii. **N-ary Tree (Generic Tree)**

An N-ary tree is a generalization of a binary tree, such that each node can have at most N children.

E**xamples of N-ary Trees:**

- **B-tree:** A self-balancing search tree where nodes can have multiple children, usually used for indexing large databases.
- **B+ Tree**: A B+ treeis a variation of the B-tree that only stores data in the leaf nodes, making range queries more efficient.
- **Trie (Prefix Tree)**: A tree where each node represents a character, and paths from the root to leaves represent strings. It can have a variable number of children for each node, making it an N-ary tree.

Consider the tree below